COOLSHADE · FEASIBILITY ASSESSMENT

CoolShade feasibility assessment

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DOCUMENT CONTROL

Document ID	CSH-FEA-001	Version	0.1
Status	Draft	Date	2026-10-30
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/08-feasibility.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	First issue in the kit 1.9.1 upgrade, from CSH-STM-001, CSH-CAL-001 v0.6 and the D3 model of option A

CoolShade feasibility assessment

Feasible on paper, with three open questions that only a site and maker's data can close: the bolted frame, the energy budget and the cooling pass their checks with margin (CSH-CAL-001, CSH-STM-001), but the droplets must evaporate before they wet anyone (STEM S4), the water must stay safe (S14) and the cost, USD 882, is 13.8 % over the USD 775 target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	A 3.0 x 2.4 m shade at about 7 degrees on four 80 x 80 x 3 mm posts; the mist absorbs about 18.8 kW and cools the zone 2.7 °C while spraying; 85 L and 204 Wh per design day; the D3 model builds with 372 parts and no interference	High	Real mixing is less even than the control volume assumes (S2); wetting needs Dv0.9 below about 80 µm at 7 bar (S4)	CSH-STM-001 S1 to S10; CSH-CAL-001 sections A to F; `cad/fidelity.json`
Materials	Galvanised steel sections, HDPE shade cloth, an aluminium panel frame, polyethylene tank; brass nozzles	High	Scale in hard water blocks the 0.4 mm nozzles (S12); 90 % UV-block cloth is untested for the radiant benefit	CSH-STM-001 S12, S13
Manufacturing and assembly	Cut, drill, bolt; M8 rivet nuts in 2 mm tube; no welding; two people in two days on paper	Medium	Needs a hand rivet nut tool and a vice (CSH-DEC-001); the rivet nut pull-out rating in a 2 mm wall is unchecked (S21); build time unverified	CSH-BLD-001; CSH-CAL-001 G15, H2
Power and controls	100 W panel, 25 Ah LiFePO4, MPPT, microcontroller, 12 V pump; 1.66 times the daily use harvested; 1.26 design days of reserve	High	Charging pauses in air above about 42.7 °C with the white enclosure (S11); a PIR sensor loses range in air near body temperature (S15); firmware is TRL 4 scope	CSH-CAL-001 F1 to F5; CSH-STM-001 S9, S10, S11, S15
Supply of parts	Steel sections, cloth, panel, battery, pump, tank and nozzles from regional distributors; BOM of 16 priced lines	Medium	Steel prices look low until quoted (CSH-CAL-001 J2); nozzle flow and droplet data not yet from a maker; USD 882 against USD 775	`bom/bom.csv`; CSH-DEC-001 to-confirm items 1 to 3
Site	An existing concrete slab for M16 anchors; permission from the asset owner; a hygiene plan accepted by the health authority; someone to refill and drain daily	Low	No partner or site agreed (CSH-DEC-001); the hygiene plan is a pre-condition of a public trial (CSH-REG-001 G1)	CSH-REG-001 G1, G2; CSH-RSK-001 P1, P3

Ratings: High, Medium, Low.